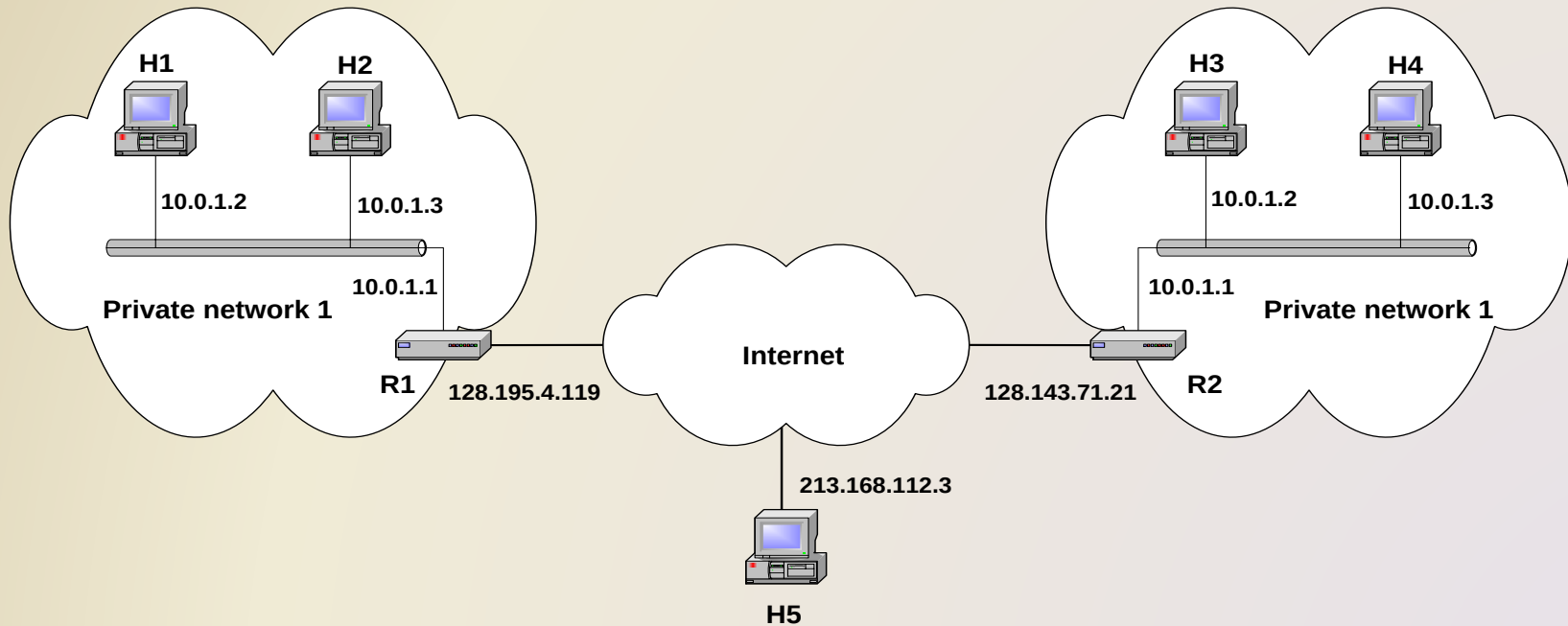


Network Address Translation

Computer Networks

Network Address Translation (NAT)

- NAT (Network Address Translation) Maps Private IPs to Public IPs
- It is required because of shortage of IPv4 Address



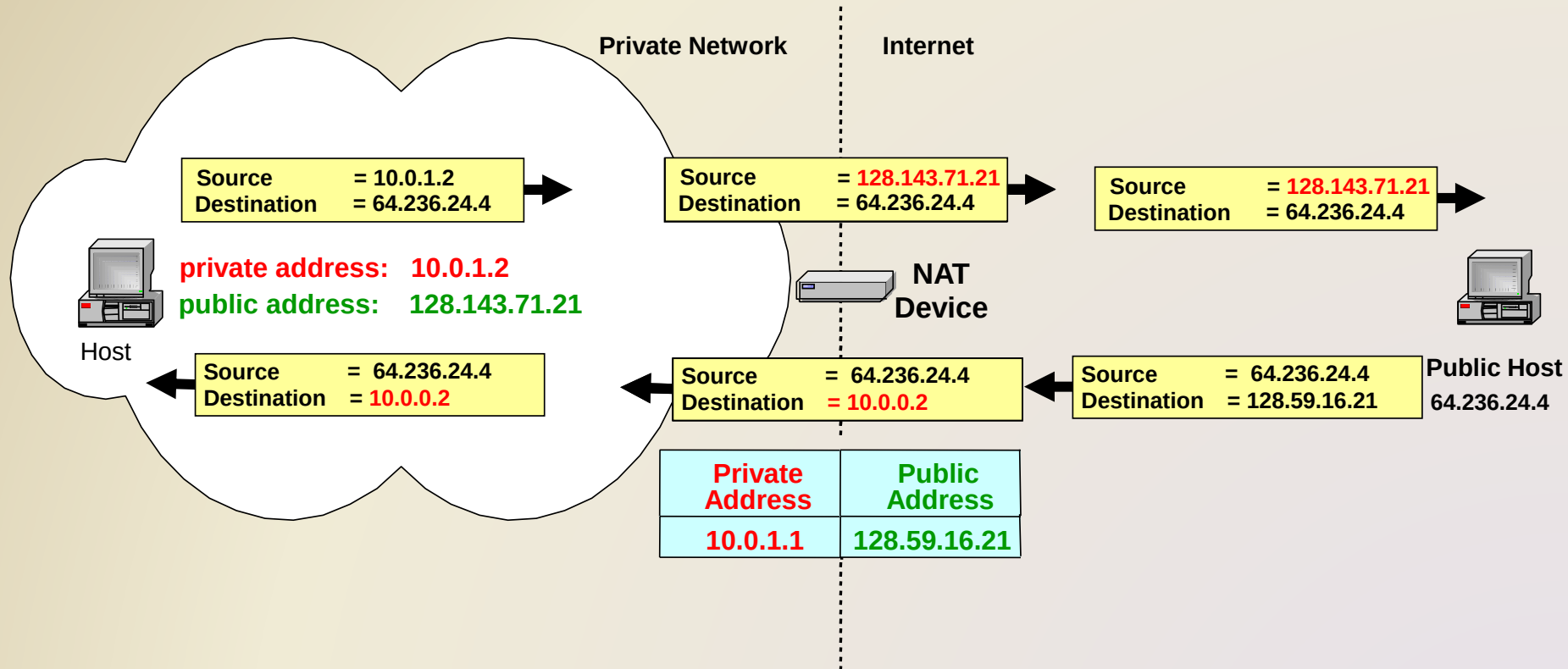
Private and Public IP Addresses

- Whatever connects directly into Internet must have public (globally unique) IP address
- There is a shortage of public IPv4 address
- So Private IP addresses can be used within a private network
- A private IP is mapped to a Public IP, when the machine has to access the Internet

Network Address Translation (NAT)

- A short term solution to the problem of the deficiency of IP addresses
 - Long term solution is IPv6
 - CIDR (Classless Inter Domain Routing) is a possible short term solution
 - NAT is another
- NAT is a way to conserve IP addresses
 - Can be used to hide a number of hosts behind a single IP address
- Three address ranges are reserved for private usage
 - 10.0.0.0-10.255.255.255,
 - 172.16.0.0-172.32.255.255 or
 - 192.168.0.0-192.168.255.255

Basic Operation of NAT



- NAT device stores the IP address and port translation tables
 - In this example, mapped only IP addresses.